A Machine-Learning-Based Approach to Expedite Predicting Effective Metamaterial Properties for Electromagnetic Transmission Control

Sarawar Hossain*, Md. Tasbiul Karim Tahsin†, Yakub Ali‡, Prof. Mohammed Shamsul Alam§, Dr. Touhidul Alam¶
*†‡§¶ International Islamic University Chittagong

***Abstract*—Metamaterials are engineered structures capable of manipulating electromagnetic (EM) waves in unconventional ways, enabling applications in EM shielding, cloaking, and wave absorption. However, their design process is typically computationally intensive and reliant on iterative simulations. This paper presents a machine learning (ML)-based framework to accelerate the prediction and design of metamaterials with desired EM properties. A deep learning model is trained to map geometric and material parameters to EM characteristics such as permittivity, permeability, and scattering parameters. A dataset of over 1,100 simulated metamaterial samples was generated using CST Studio Suite across the 20–32 GHz frequency range through an automated Python interface. A backward prediction model based on a custom artificial neural network (ANN) architecture is also introduced to generate metamaterial designs from target frequency responses. The proposed approach significantly reduces design time while maintaining high predictive accuracy, demonstrating the potential of ML-driven metamaterial design for next-generation 5G and radar applications.**

I. Introduction

The management of electromagnetic (EM) radiation has become increasingly important in modern technologies such as wireless communication, radar, and electronic systems. Conventional materials like conductive metals and ferrites are widely used for EM shielding; however, they often suffer from drawbacks including high weight, limited flexibility, and narrowband operation. These challenges have motivated the development of advanced artificial structures known as metamaterials, which can manipulate EM waves through subwavelength geometries rather than intrinsic material composition.

Metamaterials enable unprecedented control over wave propagation, reflection, and absorption, offering potential for applications in stealth, cloaking, and EM compatibility. Despite their promise, designing metamaterials with specific electromagnetic responses remains a computationally expensive task. Traditional simulation techniques such as the finite element method (FEM) and finite-difference time-domain (FDTD) provide accurate results but require extensive iterative optimization, making large-scale exploration of design spaces impractical.

Recent advances in machine learning (ML) have introduced data-driven alternatives for accelerating metamaterial design. ML models can learn the nonlinear relationships between geometric parameters and EM responses, enabling rapid prediction and optimization without exhaustive simulations. Prior studies have demonstrated progress using neural networks and ensemble learning methods, yet challenges persist regarding generalization across frequency bands, limited dataset diversity, and the inverse prediction of optimal structures from desired EM responses.

To address these issues, this paper proposes a unified ML-based framework for both forward and inverse prediction of metamaterial properties. A comprehensive dataset was generated using CST Studio Suite through an automated Python interface, capturing over 1,100 unique designs within the 20–32 GHz frequency range. Deep learning models were trained to predict key EM parameters, while a custom inverse network was developed to generate candidate designs for target frequency responses. The proposed approach reduces design time and computational cost while maintaining high accuracy, facilitating faster development of EM shielding and absorption structures for next-generation communication and sensing applications.

Contributions

This work presents a machine learning framework for metamaterial design, enabling both forward prediction of electromagnetic responses from design parameters and inverse prediction of structural parameters from desired responses. A large-scale, automated dataset was generated using CST Studio Suite over the 20–32 GHz frequency range, providing a reliable foundation for model training. The framework employs a deep neural network achieving high predictive accuracy while significantly reducing computation time. The proposed approach demonstrates the practical potential of ML-enabled metamaterial design for applications in electromagnetic shielding, radar systems, and 5G communication technologies.

II. Related Work

The integration of machine learning (ML) into metamaterial research has transformed electromagnetic (EM) design workflows, enabling faster and data-driven approaches compared to traditional simulation-based methods. ML techniques allow the exploration of complex, high-dimensional parameter spaces while significantly reducing computational costs. These capabilities are essential for applications such as EM shielding, stealth, wireless communication, and biomedical sensing.

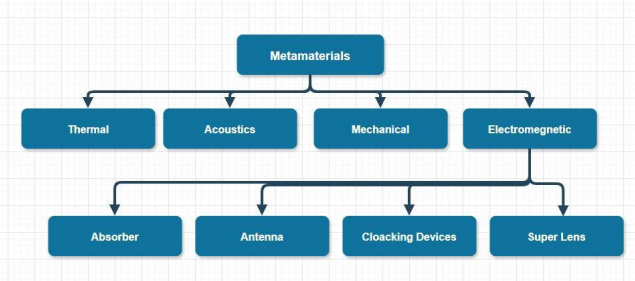


Fig. 1. Types of metamaterials and their applications in electromagnetic domains.

TABLE I
SELECTED STUDIES ON AI-DRIVEN METAMATERIAL DESIGN.

Research Focus	Methods	Key Contributions
Sun *et al.* [21]: NZIM optimization for antenna design	Elite-preserving GA	Achieved high-directivity antennas using near-zero index metamaterials
Mohammed *et al.* [12]: 5G metamaterial absorber design	MMA design	Developed compact, five-band absorber with near-perfect performance across 20–26 GHz
Cerniauskas *et al.* [1]: Data-driven optimization	ANN, GA, PCA	Introduced dimensionality reduction for efficient metamaterial design exploration
Qian *et al.* [15]: Intelligent metasurface design	Tandem NN, CNN	Demonstrated ML-enabled metasurface design for advanced EM applications
Si *et al.* [18]: AI-assisted metamaterial wave control	CNNs, ML models	Highlighted AI-based forward and inverse prediction in plasmonic systems
Chen *et al.* [3]: Interpretable inverse design	Random-forest RIGID	Proposed a fast, interpretable inverse design model with minimal data needs
Lee *et al.* [6]: Structural metamaterial optimization	FEA, ML generation	Achieved 90% higher buckling strength through ML-based generative design

Despite these advances, challenges remain in scalability, interpretability, and data efficiency. Hybrid or physics-informed ML models are increasingly investigated to improve prediction reliability and physical consistency. Traditional electromagnetic design methods, including the finite element method (FEM) and finite-difference time-domain (FDTD) simulations, provide high accuracy but are computationally expensive and unsuitable for rapid design iteration. Heuristic algorithms such as genetic algorithms (GA), particle swarm optimization (PSO), and topology optimization have improved efficiency but still depend on extensive iterative simulations.

Recent studies have demonstrated ML's strength in both forward and inverse metamaterial design. Peurifoy *et al.* [14] used deep neural networks (DNNs) for predicting optical responses of multilayer nanostructures, achieving orders-of-magnitude faster performance than conventional solvers. Liu *et al.* [10] employed convolutional neural networks (CNNs) for efficient feature extraction of EM responses. Ensemble models such as random forests (RF) and Gaussian process regression (GPR) have also been utilized to quantify uncertainty in surrogate predictions.

For inverse design, generative models such as variational autoencoders (VAEs) and generative adversarial networks (GANs) have become popular. Zhang *et al.* [23] proposed a GAN-based framework that generated novel metamaterial geometries with superior absorption performance. Hybrid designs combining GANs with physics constraints and evolutionary optimization have shown potential for stable and interpretable inverse prediction.

Table I summarizes key recent works applying AI and ML to metamaterial design. These contributions span forward modeling, inverse generation, and interpretable hybrid frameworks. The overall trend emphasizes generalization across frequencies, physics-informed learning, and reduced data requirements.

Overall, the literature illustrates steady progress in integrating ML into metamaterial modeling and optimization. However, several gaps persist: (1) limited generalization across wide frequency bands; (2) dependence on large, simulated datasets; (3) lack of interpretability in deep learning predictions; and (4) insufficient experimental validation of AI-generated designs. Addressing these issues motivates this work, which develops a comprehensive ML-based framework for efficient prediction and inverse design of metamaterials within the 20–32 GHz frequency range.

III. METHODOLOGY

This study presents an integrated framework combining electromagnetic (EM) simulation, automated data generation, and machine learning (ML) modeling for metamaterial design at 28 GHz. The workflow couples CST Microwave Studio with Python scripting to accelerate dataset creation and predictive modeling.

A. Research Design and Workflow

Fifth-generation (5G) communication systems rely on the millimeter-wave spectrum, particularly around 28 GHz. The metamaterial unit cell was designed to resonate near this frequency and validated over a 20–32 GHz range for stable broadband behavior.

The complete workflow was fully automated using a Python API. Geometric parameters and design templates were first defined in Python, after which CST was launched via a COM-based interface to generate unit-cell models. Frequency-domain simulations were conducted over the 20–32 GHz range using Floquet ports, and the resulting S-parameters (S_{11} and S_{21}) were extracted and stored in an SQLite database. The dataset was then preprocessed for machine learning, and models for both forward (parameter-to-response) and backward (response-to-parameter) predictions were trained and evaluated.

B. Tools and Technologies

The main simulation and automation components included CST Microwave Studio for electromagnetic design and frequency-domain simulations, Python for data generation, automation, and machine learning processing, and the `pywin32` library to enable COM communication with CST. A custom Python API automated geometry creation, boundary setup, and data extraction, while standard libraries such as `numpy`, `matplotlib`, and `sqlite3` facilitated data handling and visualization. Machine learning frameworks were employed to implement regression and deep learning models for both forward and inverse metamaterial predictions.

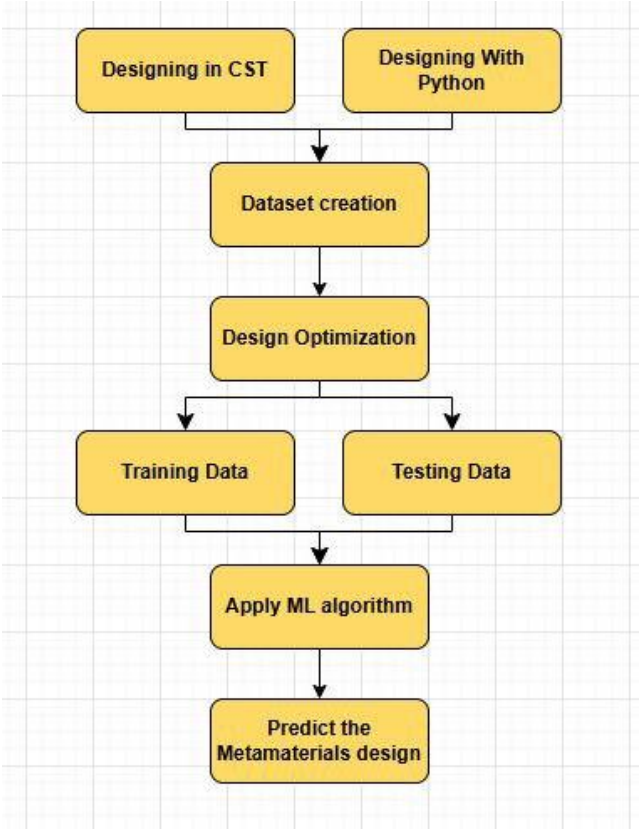


Fig. 2. Overall methodology workflow integrating CST simulation with automated dataset generation and ML training.

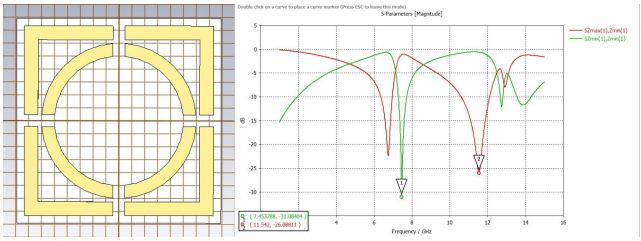


Fig. 3. Example 1: Unit-cell geometry and S-parameter response.

C. Dataset Generation

The dataset consisted of over 1,100 metamaterial structures simulated within the 20–32 GHz band. Each record contained geometric parameters and corresponding S-parameter responses. Figure 3 illustrate representative examples of the dataset, showing how geometry variations affect transmission characteristics.

D. Machine Learning Framework

Two predictive models were developed:

Forward Model: Predicts EM responses ($|S_{11}|$, $|S_{21}|$) from geometric parameters using a Light Gradient Boosting Machine (LGBM) regressor, selected for its speed and generalization.

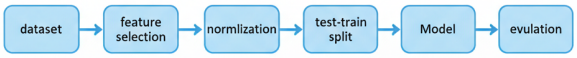


Fig. 4. Workflow of forward and backward model training.

Backward Model: Predicts geometric parameters from desired EM responses using a feedforward neural network with ReLU activation and multiple hidden layers.

E. Simulation Setup and Parameters

The EM simulations employed periodic boundary conditions in the x–y plane and open boundaries in z. The frequency-domain solver was configured for 20–32 GHz with TE(0,0) excitation. Key geometric parameters included substrate size (`ls`, `hs`), copper thickness (`t`), global cut (`gc`), and resonator dimensions (`ed1`, `id1`, `ed2`, `id2`, `l2`, `w1`).
All simulations were automatically validated to ensure convergence and correct output generation. Invalid or incomplete runs were excluded to preserve dataset integrity.

F. Validation and Limitations

Automated validation ensured accurate and consistent simulation results, enabling large-scale data generation without manual intervention. Despite this, several limitations were noted: occasional communication errors between Python and CST during long batch runs, the dataset being restricted to variations of a single topology limiting model generalization, simulations confined to the 20–32 GHz frequency band, and high computational demands that constrained the number of samples. Nonetheless, this methodology establishes a reproducible framework for metamaterial data generation and ML-based modeling. The integration of CST simulation, Python automation, and ML prediction facilitates rapid forward and inverse design, significantly reducing development time while maintaining physical accuracy.

IV. Dataset

Most existing metamaterial datasets are limited in frequency range, structure, or scale, often focusing on narrow bands such as 23 GHz or 26 GHz with a single geometry [9], [22], [24]. Many were created manually, leading to inconsistencies and limited scalability. To overcome these limitations, we developed a high-quality, automated dataset covering 20–32 GHz using CST Microwave Studio controlled through Python scripting. The dataset follows a structured, relational SQLite format to ensure reproducibility and accessibility, providing a reliable foundation for data-driven metamaterial research [7], [11].
The pipeline automatically created parameterized CST models, defined materials (FR-4 and copper), applied periodic boundary conditions with Floquet ports, ran frequency-domain simulations (20–32 GHz), extracted S_{11} and S_{21} responses, and stored both design parameters and results in an SQLite

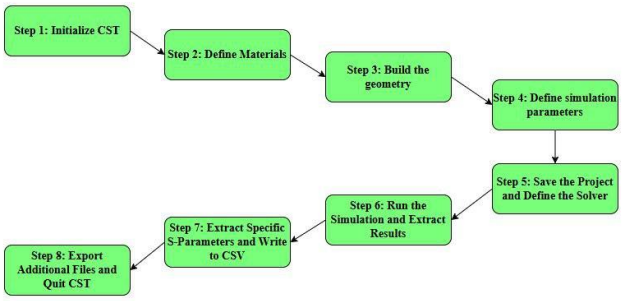


Fig. 5. Automated dataset generation methodology integrating CST with Python scripting.

TABLE II
DESIGN PARAMETERS AND SIMULATION OUTPUTS.

Category	Parameter	Description
Design Parameters	`ls`	Substrate length
	`hs`	Substrate height
	`t`	Copper thickness
	`gc`	Global cut width
	`l1`, `w1`	Outer box dimensions
	`ed1`, `id1`	Outer circle radii
	`l2`	Inner box dimension
	`ed2`, `id2`	Inner circle radii
Simulation Outputs	`s11_freq`	Frequency (GHz) of min S_{11}
	`s11_db`	Minimum reflection (dB)
	`s21_freq`	Frequency (GHz) of min S_{21}
	`s21_db`	Minimum transmission (dB)

database. Over 1,100 simulations were completed, minimizing human intervention and ensuring consistency.

Metamaterial behavior depends strongly on geometry and substrate properties. Even small dimensional changes can shift resonance frequencies and alter S-parameter responses [13], [16], [19]. These unit cells behave as LC resonators, where the inductance (L) arises from metallic paths and capacitance (C) from narrow gaps, with the resonance frequency following $f_r = \frac{1}{2\pi\sqrt{LC}}$. The FR-4 substrate and copper conductor combination effectively supports this behavior in the 20–32 GHz range.

Each dataset entry represents a unique unit-cell configuration with its corresponding electromagnetic (EM) response. Table II summarizes all input parameters and output variables. Parameters such as `gc`, `ed1`, and `id2` influence resonance frequency, while substrate parameters (`ls`, `hs`) affect bandwidth and impedance.

The dataset spans 20–32 GHz and includes 11 geometric parameters with 4 corresponding electromagnetic responses. Figure 8 illustrates representative samples, showing the organized format that enables model-ready analysis.

Exploratory correlation analysis revealed strong relationships between certain geometric parameters and resonance behavior. For instance, `gc`, `l1`, and `ed1` showed strong influence on S_{11}, while `l2` and `id2` affected S_{21} response frequencies, as visualized in Figs. **??** and **??** [8].

Table III benchmarks our dataset and design approach against

TABLE III
COMPARISON BETWEEN THE PROPOSED DESIGN AND EXISTING 5G ANTENNAS.

Ref.	Year	Material	Size (mm^2)	Res. Freq. (GHz)	Gain (dBi)	BW (MHz)
[5]	2008	FR-4	60×100	1.03, 2.13	0.66, 1.74	980, 2010
[17]	2020	RT5880	20.2×22.6	28.46	8.71	4470
[20]	2022	Kapton	5.1×5.1	28	6.46	1680
[4]	2023	FR-4	20×20	18.57, 25.24	4.58, 3.00	2000, 6720
Proposed	2025	FR-4	10×10	20–32	–	–

TABLE IV
OPTIMAL LGBMREGRESSOR PARAMETERS.

Parameter	Value
Number of estimators	1000
Learning rate	0.01
Number of leaves	31
Minimum child samples	20
Maximum depth	5
Subsample ratio	0.8
Column sample by tree	0.8
Force row-wise	True
Random state	42
Maximum bin	30

other 5G-compatible metamaterial and antenna structures. The proposed design achieves compactness and scalability while maintaining electromagnetic consistency.

The dataset's quality was verified through deterministic solvers, ensuring noise-free, reproducible outputs. Automated extraction prevented missing data, maintaining complete integrity. Although computationally intensive—each CST run required significant CPU and RAM resources—the automated pipeline enabled scalable dataset generation with minimal human supervision.

Overall, this dataset offers a comprehensive, physics-consistent foundation for both forward and inverse machine learning tasks in metamaterial design [2], [9].

V. RESULTS AND DISCUSSION

This section presents the outcomes of both forward and backward machine learning models trained on the metamaterial dataset described earlier. Two predictive tasks were explored: (1) predicting the resonance frequency (S_{21_freq}) from design parameters (forward), and (2) predicting the geometric parameters from resonance frequency (backward).

Forward Prediction (Parameters → Frequency): The forward prediction problem was treated as a regression task using several models, with the Light Gradient Boosting Machine (LGBMRegressor) achieving the best performance after testing approximately 32,000 hyperparameter combinations. The optimal parameters are listed in Table IV.

The model achieved a Root Mean Squared Error (RMSE) of **0.7225** and a Coefficient of Determination (R^2) of **0.9156**, indicating high predictive fidelity. Figure 9 shows the actual

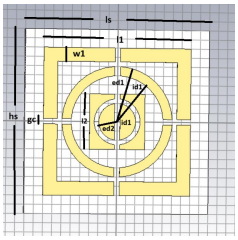


Fig. 6. Labeled diagram of metamaterial design parameters.

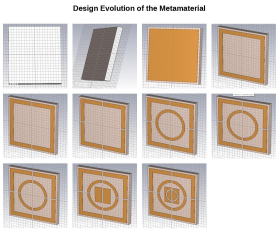


Fig. 7. Step-by-step design evolution of the metamaterial unit cell.

ls	hs	t	gc	w1	l1	ed1	id1	l2	ed2	id2	s11_freq	s11_db	s21_freq	s21_db
9.0	1.8	0.040	0.7	1.0	7.15	6.15	4.85	3.2	2.6	1.8	30.992	-29.140071	30.860	-8.973546
9.0	1.8	0.040	0.7	1.0	7.50	6.15	4.85	3.0	2.6	1.8	27.068	-36.442460	25.772	-9.974330
8.0	1.8	0.040	0.7	1.0	7.48	6.00	5.00	3.0	2.5	2.0	26.216	-17.069927	30.800	-15.962313
10.0	1.6	0.045	0.8	1.2	8.87	6.00	5.00	3.0	2.5	2.0	26.768	-30.631650	25.568	-13.141540
9.0	1.8	0.040	0.7	1.0	7.39	6.15	4.85	3.0	2.6	1.8	27.032	-36.303212	25.700	-11.133763
10.0	1.6	0.045	0.8	1.2	9.30	6.00	5.00	3.0	2.5	2.0	24.092	-19.192183	25.376	-9.544446
8.0	1.6	0.035	0.5	1.0	7.47	6.00	5.00	3.0	2.5	2.0	30.644	-32.076568	28.520	-16.534866
10.0	1.8	0.040	0.7	1.5	9.85	6.00	5.00	3.0	2.5	2.0	22.940	-33.681590	25.568	-13.564228
10.0	1.6	0.045	0.8	1.2	9.40	6.00	5.00	3.0	2.5	2.0	28.052	-26.986731	25.532	-11.427953
8.0	1.6	0.035	0.3	1.0	7.60	6.00	5.00	3.0	2.5	2.0	30.452	-18.299907	31.508	-17.135965

Fig. 8. Sample entries from the metamaterial dataset.

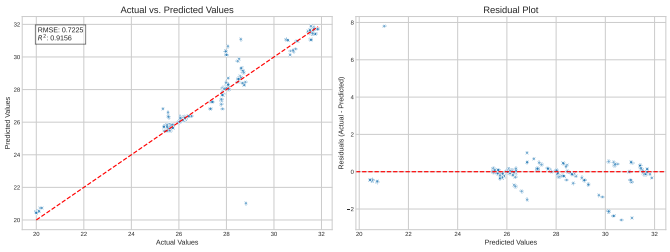


Fig. 9. LGBMRegressor performance: (a) Actual vs. Predicted values, (b) Residuals.

TABLE V
FORWARD PREDICTION MODEL COMPARISON (TARGET: S_{21_FREQ}).

Model	RMSE	R^2	Notes
LGBMRegressor	**0.7225**	**0.9156**	Best performing
Random Forest Regressor	1.0342	0.8611	Stable, less accurate
CatBoost Regressor	0.9817	0.8743	Underfitted slightly
AdaBoost Regressor	1.2764	0.8124	Poor complexity handling
Linear Regression	1.8547	0.6910	Weak nonlinear capture

versus predicted resonance frequencies and residual distribution, both confirming strong model generalization and well-behaved errors.

A comparative summary of all forward prediction models is shown in Table V. The LGBMRegressor consistently outperformed others, demonstrating robustness and generalization on unseen data.

Backward Prediction (Frequency → Parameters): The inverse prediction task aimed to reconstruct the metamaterial geometry from a given resonance frequency. A fully connected feedforward Artificial Neural Network (ANN) was implemented using `PyTorch`, mapping one input (frequency) to eleven geometric parameters. The proposed architecture follows an encoder–decoder form with dimensions 1–64–128–256–128–64–11, using ReLU activations and the Adam optimizer (lr $= 10^{-3}$). The model was trained for 50 epochs using a batch size of 32 and MSE loss.

Table VI lists the ANN architecture variants tested for ablation study.

Training convergence was smooth, reaching a final MSE loss of **0.8433**. Figure 10 illustrates the decreasing loss curve over epochs.

Predicted versus actual outputs for all eleven design parame-

TABLE VI
ANN MODEL ARCHITECTURES FOR BACKWARD PREDICTION.

Model	Layers	Activation / Reg.	Key Characteristics
Custom ANN (ours)	6	ReLU	Deep symmetric; balanced depth and expressiveness
Variant A	4	ReLU	Shallow; faster, lower capacity
Variant B	5	LeakyReLU	Better gradient flow, mild underfit
Variant C	6	ReLU + Dropout(0.3)	Reduced overfitting; higher variance
Variant D	6	ReLU (skip)	Residual-style; unstable after 40 epochs
Variant E	6	Tanh	Slow learning; gradient saturation

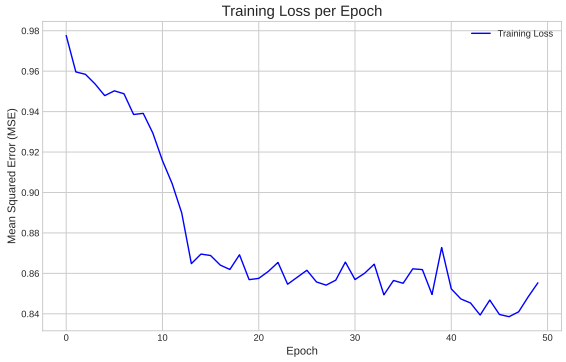


Fig. 10. Training loss curve for the proposed ANN model.

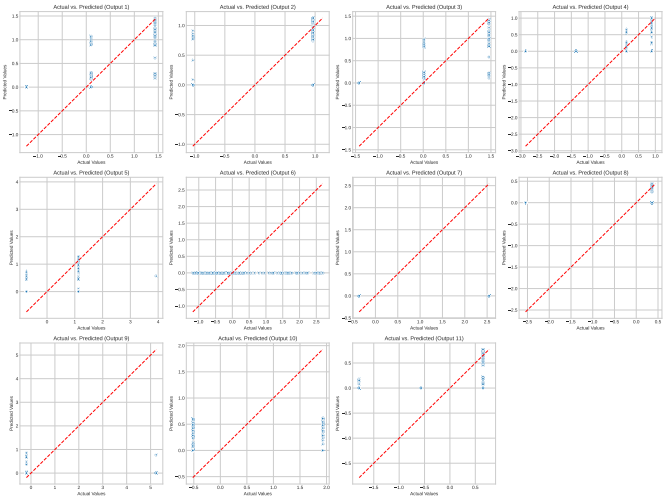


Fig. 11. Predicted vs. actual values for all eleven geometric parameters.

TABLE VII
BACKWARD PREDICTION MODEL COMPARISON (LOSS AFTER 50 EPOCHS).

Model	Epochs	Final Loss	Notes
Custom ANN (ours)	**50**	**0.8433**	Best performance
ANN Variant 1	50	1.1274	Simpler architecture
ANN Variant 2	50	1.2841	Shallow network
ANN Variant 3	50	1.4529	Over-regularized
ANN Variant 4	50	1.3098	Underfitted

ters are shown in Figure 11. High-sensitivity parameters such as `gc` and `ed1` exhibit particularly close agreement, confirming the ANN's ability to capture nonlinear dependencies in the dataset.

Table VII summarizes the performance of all ANN variants. The proposed deep symmetric model achieved the lowest final loss, outperforming others that either underfit or over-regularized.

Both forward and backward prediction experiments demonstrate that data-driven methods can effectively model complex EM behavior. Gradient boosting (LGBM) provided high accuracy with low variance, making it ideal for parametric prediction tasks. The ANN successfully reconstructed geometric features from resonance frequencies despite the one-to-many mapping challenge.

These findings emphasize that proper model architecture, depth, and nonlinearity are essential for reliable metamaterial design. The combination of physics-based data generation and ML-based prediction enables faster exploration of large design spaces, paving the way for automated metamaterial design frameworks.

VI. Conclusion and Future Work

A. Conclusion

This work presented a machine learning-based framework for predicting and designing electromagnetic (EM) metamaterials, focusing on transmission control within the 20–32 GHz range. A fully automated pipeline was developed by integrating CST Microwave Studio with Python, enabling automatic generation, simulation, and data extraction of metamaterial unit cells without manual intervention.

A comprehensive dataset was constructed, capturing the electromagnetic response characteristics required for model training. The Light Gradient Boosting Machine (LGBM) achieved the highest accuracy for forward prediction of resonance frequency, with an R^2 score of 0.9156 and RMSE of 0.7225. For inverse prediction, a custom Artificial Neural Network (ANN) demonstrated robust performance with a final mean squared error of 0.8433 after 50 epochs.

The results confirm that machine learning can significantly accelerate the metamaterial design process by reducing the dependency on computationally expensive simulations. The proposed framework offers a scalable approach for efficient EM property prediction, supporting faster iterations and data-driven optimization of metamaterial structures. These findings establish a strong foundation for future advancements in automated metamaterial design and electromagnetic shielding applications.

B. Future Work

Several directions can further extend this research:

Enhanced CST–Python Integration: Expanding the automated interface to extract additional simulation outputs such as electric field distributions and effective material parameters will enrich the dataset and improve model interpretability.

Geometric Diversity: Incorporating a wider variety of unit-cell geometries will improve generalization and enable models to handle more complex or novel metamaterial structures.

Dataset Expansion: Increasing dataset size and extending the frequency range beyond 32 GHz will allow the development of deeper learning architectures and enhance prediction reliability across broader electromagnetic bands.

Optimization Integration: Coupling trained models with optimization algorithms (e.g., Bayesian or genetic search) can facilitate real-time inverse design and performance tuning for specific applications.

By addressing these areas, the integration of machine learning into metamaterial design can become a comprehensive tool for rapid, accurate, and scalable EM material innovation.

References

[1] G. Cerniauskas, H. Sadia, and P. Alam. Machine intelligence in metamaterials design: a review. *Oxford Open Materials Science*, 4(1):itae001, 2024.

[2] Mengzhou Chen, Shuang Hong, Jinhui Kim, Demetrios L Sounas, Andrea Alù, and Xiang Zhang. Inverse design of metastructures using machine learning. *Nature Communications*, 11(1):1–9, 2020.

[3] W. Chen, R. Sun, D. Lee, C.M. Portela, and W. Chen. Generative inverse design of metamaterials with functional responses by interpretable learning. *Advanced Intelligent Systems*, page 2400611, 2024.

[4] Namira Chowdhury, Humaira Binte Rahman, Sikder Sunbeam Islam, Mohammad Shamimul Haque Choudhury, Mandeep Jit Singh, Norbahiah Misran, Mohammad Jakir Hossain, and Mohammad Tariqul Islam. A split square-shaped metamaterial loaded microstrip patch antenna for 5g communication. *Journal of Advanced Research in Micro and Nano Engineering*, 21(1):41–53, 2024.

[5] Ajay Gummalla, Cheng-Jung Lee, and Maha Achour. Compact metamaterial quad-band antenna for mobile application. In *2008 IEEE Antennas and Propagation Society International Symposium*, pages 1–4. IEEE, 2008.

[6] S. Lee, J. Kwon, H. Kim, R.O. Ritchie, and G.X. Gu. Advancing programmable metamaterials through machine learning-driven buckling strength optimization. *Current Opinion in Solid State and Materials Science*, 31:101161, 2024.

[7] X Li, Y Zhang, H Wang, and M Chen. Automatic generation of electromagnetic datasets for metamaterial prediction. *IEEE Access*, 9, 2021.

[8] D Liu, W Zhang, and H Chen. A survey of surrogate models for electromagnetic design optimization. *Electromagnetics*, 40(2), 2020.

[9] Yong Liu and Hong Chen. Machine learning for metamaterials: a review. *Journal of Materials Chemistry C*, 8(26):8774–8791, 2020.

[10] Zhichao Liu, Dongyao Zhu, Zixuan Liu, Dongyao Zhu, and Zixuan Liu. A generative model design approach for plasmonic metasurfaces by using artificial neural networks. *Scientific Reports*, 8(1):1–9, 2018.

[11] Wei Ma, Fei Cheng, Yue Jiang, Junhong Xu, Yuan Xu, Xiaojun Liang, Sijie Xiao, Yu Huang, Chang-Won Lee, and Bo Li. Deep learning for the design of photonic and plasmonic devices. *ACS Photonics*, 5(8):3458–3471, 2018.

[12] S.A. Mohammed, R.A.K. Albadri, and K.S.L. Al-Badri. Simulation of the microwave five-band a perfect metamaterial absorber for the 5g communication. *Heliyon*, 9(9), 2023.

[13] M. Moniruzzaman et al. Electromagnetic characterization of mirror symmetric resonator based metamaterial and frequency tuning: a dielectric based multilayer approach. *Microwave and Optical Technology Letters*, 2022.

[14] Jonathan Peurifoy, Danny Vercruysse, Owen D Miller, Steven G Johnson, Jonathan Peurifoy, Danny Vercruysse, Owen D Miller, and Steven G Johnson. Nanophotonic particle simulation and inverse design using artificial neural networks. *Science Advances*, 4(6):eaar4206, 2018.

[15] C. Qian, I. Kaminer, and H. Chen. A guidance to intelligent metamaterials and metamaterials intelligence. *Nature Communications*, 16(1):1154, 2025.

[16] R. Ramachandran and et al. Symmetric square shaped metamaterial structure with quintuple resonance frequencies for s, c, x and ku band applications. *International Journal of RF and Microwave Computer-Aided Engineering*, 2023.

[17] Efri Sandi, Aodah Diamah Rusmono, Diamah Aodah, and Karisma Vinda. Ultra-wideband microstrip array antenna for 5g millimeter-wave applications. *Journal of Communications*, 15(2):198–204, 2020.

[18] L. Si, R. Niu, C. Dang, X. Bao, Y. Zhuang, and W. Zhu. Advances in artificial intelligence for artificial metamaterials. *APL Materials*, 12(12), 2024.

[19] S. A. Sifat et al. Development of double-layer metamaterial with high effective medium ratio values for s- and c-band applications. *Results in Physics*, 2021.

[20] D. Subitha, S. Velmurugan, and S. Balasubramani. Slotted square microstrip patch antenna for 5g communication at 28 ghz with improved bw and gain. In *AIP Conference Proceedings*, volume 2405. AIP Publishing, 2022.

[21] S. Sun, Y. Jiang, and J. Wang. Automatic design of pixelated near-zero refractive index metamaterials based on elite-preserving genetic algorithm optimization. *Results in Physics*, 48:106461, 2023.

[22] Hongjun Wang, Zhijian Chen, Siyang Yu, Feng Long, Jianye Li, Longfei Li, Jian Liu, and Yulin Hao. A review on machine learning methods for metamaterial design. *Materials & Design*, 216:110590, 2022.

[23] Bo Zhang, Junwei Liu, Xiaotian Chen, Hong Yang, Jian Wang, and Jun Li. Gan-based inverse design of complex metamaterial configurations with superior absorption characteristics. *Optics Express*, 29(10):15591–15600, 2021.

[24] Yuntian Zhang, Junying Liu, Min Yang, Zichang Li, Xuetao Chen, Ping Lu, Min Zhang, Jianping Yang, and Hong-Bo Sun. Deep-learning-enabled rapid design of electromagnetic metasurfaces. *Advanced Science*, 8(1):2003734, 2021.